



Briefing note

How does a carbon price translate into the price of fuel?

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The M-Token poster makes the claim that upstream carbon pricing is never likely to rise high enough to strongly influence individual mobility choices. It shows that a €100 per tonne increase in the upstream carbon dioxide equivalent price adds only about €0.24 per litre to the price of petrol. Mathematically, it can be seen that it would take a €400 per tonne increase in the upstream price of carbon to add almost €1.00 per litre to the price of petrol. The resulting increase in the retail price is relatively small, and demand for motor fuel is price-inelastic. A large increase in the upstream carbon price therefore produces a proportionately smaller reduction in fuel consumption.

The arithmetic

A fuel's emissions factor states the kilograms of CO₂-e associated with combustion of a litre of the fuel. An upstream carbon price is stated per tonne, or per 1,000 kg. The effect of a price increase is therefore:

$$\text{price increase per litre} = \text{carbon-price increase per tonne} \times \text{emissions factor} \div 1,000$$

At a carbon-price increase of 100 currency units per tonne, the relationship is especially easy to see, and it doesn't matter which currency you are working with. If the emissions factor is 2.4 kg CO₂-e/L, the price increase is 0.24 currency units/L. For an increase of 100 currency units per tonne, simply move the decimal point in the emissions factor one place to the left: 2.4 kg CO₂-e becomes 0.24 currency units per litre. At 400 currency units per tonne, it is 0.96/L, or almost 1.00/L.

What emissions factors are we talking about?

For this purpose, spurious precision is unhelpful. New Zealand's Ministry for the Environment 2026 guidance gives direct-combustion factors of 2.361 kg CO₂-e/L for regular petrol, 2.415 kg CO₂-e/L for premium petrol, and 2.672 kg CO₂-e/L for diesel. These support using rounded working values of about 2.4 for petrol and 2.7 for diesel.

Biofuel blends can have different factors. The same guidance, for example, gives lower CO₂-e factors for E10 petrol and B20 diesel than for their conventional fossil-fuel counterparts. That matters for M-Tokens because a lower prescribed emissions factor would mean fewer M-Tokens surrendered per litre.

What the number means

The calculation is currency-neutral: a 100-unit increase per tonne produces almost a quarter-unit increase per litre of petrol. The exact result varies with the fuel and the official emissions factor used. The point of the poster is the order of magnitude, not the second or third decimal place.

This does not imply that upstream carbon pricing is unnecessary or ineffective. It shows why even a very large increase in the upstream carbon price translates into a much smaller price change at the fuel pump. Combined with the price-inelasticity of demand for motor fuel, this helps explain why upstream carbon pricing may never rise high enough to strongly influence individual mobility choices. It is therefore worth investigating a complementary downstream quantity instrument for light-fleet emissions.

Sources: New Zealand Ministry for the Environment, Measuring Emissions Catalogue 2026. New Zealand Ministry of Transport, Estimation of transport-related elasticities, DTCC Annual Research 2023/24.